

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (AI&DS) (CBCS) Degree Examinations

December 2023

AD1101-1 – COMPUTER ORGANIZATION

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The decoded instruction is stored in _____.
 A) IR
 B) PC
 C) Register
 D) MDR
2. The instruction -> Add LOCA, R0 does _____.
 A) Adds the value of LOCA to R0 and stores in the temp register
 B) Adds the value of R0 to the address of LOCA
 C) Adds the values of both LOCA and R0 and stores it in R0
 D) Adds the value of LOCA with a value in accumulator and stores it in R0
3. Which registers can interact with the secondary storage?
 A) MAR
 B) PC
 C) IR
 D) R0
4. The first machine cycle of an instruction is always a
 A) Memory read cycle
 B) Fetch cycle
 C) I/O read
 D) Memory write cycle
5. A stack-based CPU organization uses _____ address instruction
 A) 2
 B) 0
 C) 1
 D) 3
6. The beginning of the architecture of the Itanium processor took place at _____.
 A) Intel
 B) Microsoft
 C) Hewlett-Packard
 D) Dell
7. Define Mapping Process
 A) It is a process of transforming data from main memory to cache memory.
 B) It is a process that signifies the validity of the locality of reference.
 C) It is a process, which translates the main memory address to the cache memory address.
 D) It is a process of detecting a word in the cache.
8. _____ is the logical structure of a computer's Random-Access Memory (RAM).
 A) Memory addressing
 B) Operation field
 C) Address field
 D) Addressing mode
9. _____ mapping is used in cache organization which is the quickest and most supple organization.
 A) set associative
 B) direct
 C) sequential
 D) associative
10. _____ address of the operand calculated during the prior cycle is used to access memory.
 A) Memory access completion cycle
 B) Instruction decode fetch cycle
 C) Instruction fetch cycle
 D) Memory access fetch cycle
11. Which of the following is the subcategories of computer architecture?
 A) Micro architecture
 B) Instruction set architecture
 C) Systems design
 D) All of the mentioned
12. Which of the architecture is power efficient?
 A) RISC
 B) ISA
 C) IANA
 D) CISC

13. To reduce the memory access time we generally make use of _____.
 A) SDRAM's B) Heaps
 C) Cache's D) Higher capacity RAM's
14. The reason for the cells to lose their state over time is _____.
 A) Use of Shift registers B) The lower voltage levels
 C) Usage of capacitors to store the charge D) Higher voltage levels
15. In a 4M-bit chip organization has a total of 19 external connections, then it has _____ address if 8 data lines are there.
 A) 2 B) 5
 C) 9 D) 8
16. During a write operation if the required block is not present in the cache then _____ occurs.
 A) Write miss B) Write latency
 C) Write hit D) Write delay
17. _____ are the different type/s of generating control signals.
 A) Hardwired B) Micro-instruction
 C) Micro-programmed D) Both Micro-programmed and Hardwired
18. The instruction, Add #45,R1 does _____.
 A) Adds the value of 45 to the address of R1 and stores 45 in that address B) Adds 45 to the value of R1 and stores it in R1
 C) Finds the memory location 45 and adds that content to that of R1 D) Finds the content of R1 and adds that content to memory location 45.
19. Which memory device is generally made of semiconductors?
 A) RAM B) Hard disk
 C) Floppy disk D) CD drive
20. Which of the following statements are correct about the main memory of a computer?
 A) In main memory, data gets stores even when power is switched off. B) Main memory is faster than secondary memory but slower than registers.
 C) They are made up of raw-conductors. D) Main memory is very slow compared to backing store

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the basic functional units of a computer with neat diagram.	4	L*2	1	1
	b) What is performance measurement? Explain over all SPEC rating for computer.	4	L2	1	1,2
	c) With the help of control sequences explain execution of the instruction Add (R3), R1 in single bus architecture.	8	L2	3	1,2,3
2.	a) List and explain the role of Processor registers during memory interaction.	4	L2	1	2
	b) Explain the importance of stacks in Subroutine nesting.	4	L2	1	1,2
	c) With the help of diagram explain control sequence for the instruction Add R4, R5, R6, in three bus organization.	8	L2	3	1,2,3
3.	a) Explain Hardwired Control unit.	4	L2	4	1,2
	b) With the help of control sequences explain the process of execution of a branch instruction in single bus architecture.	4	L2	3	1,2
	c) Define addressing mode. Propose any 4-addressing mode with proper syntax.	8	L1	1	1,2
Unit – II					
4.	a) With necessary diagram explain how Input devices can be interfaced with an I/O bus.	4	L2	5	1,2
	b) What are interrupts? Explain how nested interrupts are handled.	4	L2	5	1,2
	c) Explain restoring binary division algorithm. Perform 13/3 using restoring binary division algorithm.	8	L3	2	1,2

5.	a)	Explain the concept of Synchronous bus in data transfer operation.	4	L2	5	1,2
	b)	What is exception? Explain different types of exceptions.	4	L2	2	1,2
	c)	With the help of neat diagram explain Sequential multiplication process by multiplying 13*11.	8	L3	2	1,2
6.	a)	Multiply 13 and -6(Multiplier) using booth's bit pair recoding algorithm.	4	L3	2	1,2
	b)	What is Direct Memory Access? Explain the role of DMA controller during data transfer.	4	L2	5	1,2
	c)	Explain the following:				
	i.	Universal Serial Bus (USB)				
	ii.	Peripheral Component Interconnect (PCI) bus	8	L2	5	1,2

Unit – III

7.	a)	Explain Associative Mapping technique.	4	L1	4	1,2
	b)	Explain Address translation process in Virtual memory.	4	L2	3	1,2
	c)	List and explain in detail different cache replacement algorithms.	8	L2	4	1,2
8.	a)	Write a short note on Virtual Memory.	4	L2	4	1,2
	b)	Define ROM. Compare different types of ROM.	4	L1	4	1,2
	c)	Explain with neat diagram internal organization of 2M*8 dynamic memory chip.	8	L2	4	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
